

Algebraic Geometry

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This is a first course in modern algebraic geometry, essentially covering chapter II and selective sections of chapter III of [Har77]. I also included some results on [Har77] I found to be essential but were not included in this course due to the limited time. The examples and exercises are from either example sheets or past-year papers. The last section on Serre-duality is extremely rushed and unusable. Just go read chapter III of [Har77]

1 Sheaves

Given a topological space X , let $\mathcal{C}$ be the category of its open subsets with homomorphism $U \rightarrow V$ if $U \subset V$. A **presheaf** of abelian groups on X is a functor $\mathcal{F} : \mathcal{C}^{op} \rightarrow \mathcal{A}b$. It is a **sheaf** if for any open U and an open covering $\{U_i\}$ of U it satisfies the exact sequence.

$$0 \rightarrow \mathcal{F}(U) \rightarrow \prod_i \mathcal{F}(U_i) \xrightarrow{p_1^* - p_2^*} \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

where p_1 is the projection via the first coordinate.

Remark 1.1. If $\mathcal{F}$ is a sheaf, then $\emptyset \subset X$ is covered by the empty covering, and hence $\mathcal{F}(\emptyset) = 0$.

Example 1. (constant sheaf) Take any group G and set $\mathcal{F}(U) = G$. This is called the constant presheaf. It is not a sheaf in general which can be seen by taking two disjoint U and V . Its sheafification $\mathcal{F}^+(U) = \{f : U \rightarrow G \text{ continuous}\}$ by giving G the discrete topology (that is, the locally constant functions on U). This called the **constant sheaf** by G and now it is clear that $\mathcal{F}(U)$ is just copies of G with copy number agreed with the number of its connected component.

Let $\mathcal{F}$ be a presheaf. The **stalk** of $\mathcal{F}$ at x is

$$\mathcal{F}_x = \{(U, s) | U \text{ open nbhd of } x, s \in \mathcal{F}(U)\} / \sim$$

where $(U, s) \sim (V, s')$ at x if there exists $W \subset U \cap V$ such that $s|_W = s'|_W$. Then by definition, $\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U)$

Remark 1.2. We will think of $\mathcal{F}(U)$ as regular functions on U and $\mathcal{F}_x$ the germs at x .

Proposition 1.1. A morphism of sheaves $f : \mathcal{F} \rightarrow \mathcal{G}$ is an isomorphism (resp. injective, surjective) if and only if $f_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is an isomorphism (resp. injective, surjective) at each $x \in X$

Given a presheaf $\mathcal{F}$, there exists a sheaf $\mathcal{F}^+$ and a morphism $\theta : \mathcal{F} \rightarrow \mathcal{F}^+$ with the universal property that for any sheaf $\mathcal{G}$ and morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$, φ uniquely factors through θ . From the universal property $(\mathcal{F}^+, \theta)$ is unique up to isomorphism, and is called the **sheafification** of $\mathcal{F}$.

We will not be bothered with the construction of sheafification but only concern its universal property, which enables us to always deal with morphism of sheaves.

Lemma 1.1. $\theta_x : \mathcal{F}_x \rightarrow \mathcal{F}_x^+$ is an isomorphism

This lemma provides us with much convenience when examining a morphism at the level of stalks.

Given a morphism of presheaves $f : \mathcal{F} \rightarrow \mathcal{G}$, we get natural definition of presheaves of kernel, image and cokernel. If f is a morphism of sheaves, it turns out that the presheaf of kernel $\ker f$ is sheaf but not necessary for the two left. So we define and sheaf of image $\operatorname{Im} f$ the sheafification of the presheaf of the image of f and the same for cokernel.

We say f is injective if $\ker f = 0$ and surjective if $\operatorname{Im} f = 0$. A sequence

$$\dots \xrightarrow{f} \mathcal{F} \xrightarrow{g} \dots$$

is exact at $\mathcal{F}$ if $\operatorname{Im} f = \ker g$

Lemma 1.2. $(\ker f)_x = \ker(f_x)$, $(\operatorname{Im} f)_x = \operatorname{Im}(f_x)$

Let $f : X \rightarrow Y$ be a continuous map and $\mathcal{F}$ a sheaf on X and $\mathcal{G}$ a sheaf on Y . Define the **pushforward** of $\mathcal{F}$ the sheaf $f_*\mathcal{F}(U) = \mathcal{F}(f^{-1}(U))$ on Y , and the **inverse image presheaf** $(f^{-1}\mathcal{G})^{\text{pre}}(U) = \varinjlim_{f(U) \subset V} \mathcal{G}(V)$. The **inverse image sheaf** is the sheafification of $(f^{-1}\mathcal{G})^{\text{pre}}$.

Remark 1.3. We will see that for an $\mathcal{O}_X$ -module $\mathcal{F}$ its pushforward is an $\mathcal{O}_Y$ -module. However, if $\mathcal{G}$ is an $\mathcal{O}_Y$ -module the inverse image sheaf of $\mathcal{G}$ is not naturally an $\mathcal{O}_X$ -module but only an $f^{-1}\mathcal{O}_Y$ -module. We will define the pullback of $\mathcal{G}$ to be $f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$

Proposition 1.2. f_* and f^{-1} are adjoint functors, i.e. $\operatorname{Hom}_X(G, f_*\mathcal{F}) \cong \operatorname{Hom}_Y(f^{-1}\mathcal{G}, \mathcal{F})$

Proof. It suffices to show the existence of natural maps $f^{-1}f_*\mathcal{F} \rightarrow \mathcal{F}$ and $\mathcal{G} \rightarrow f_*f^{-1}\mathcal{G}$. For the first one, by the universal property of sheafification, it is enough to give a map $f^{-1\text{pre}}f_*\mathcal{F} \rightarrow \mathcal{F}$. Note that for U open in X , $f^{-1\text{pre}}f_*\mathcal{F}(U) = \varinjlim_{f(U) \subset V} f_*\mathcal{F}(V) = \varinjlim_{f(U) \subset V} \mathcal{F}(f^{-1}(V))$. Note that $U \subset f^{-1}(U)$ for $f(U) \subset V$, so just let $s \mapsto s|_U$ since $s \in \varinjlim_{f(U) \subset V} \mathcal{F}(f^{-1}(V))$

Similarly, for $\mathcal{G} \rightarrow f_*f^{-1}\mathcal{G}$ and U open in Y , $f_*f^{-1}\mathcal{G}(U) = \mathcal{G}(f^{-1}(U))$. Again, it is enough we give a map from $\mathcal{G}$ to the presheaf, and then by composition with θ to its sheafification we get the required map. Now $f^{-1\text{pre}}\mathcal{G}(f^{-1}(U)) = \varinjlim_{ff^{-1}(U) \subset V} \mathcal{G}(V)$. Note that $ff^{-1}(U) \subset U$, so we can just let $s \mapsto s|_{ff^{-1}(U)}$ since $s \in \varinjlim_{ff^{-1}(U) \subset V} \mathcal{G}(V)$ for $s \in \mathcal{G}(U)$ $\square$

1.1 Exercise

Our first example is **extending a sheaf by 0**. Let X be a topological space and $j : U \hookrightarrow X$ the inclusion of an open subset. Let $\mathcal{F}$ be a sheaf of abelian groups on U . Define the **extension by zero** of $\mathcal{F}$ to be the sheaf $j_!\mathcal{F}$ associated to the presheaf $j_p\mathcal{F}$ on X defined by

$$j_p\mathcal{F}(V) = \begin{cases} \mathcal{F}(V) & \text{if } V \subset U \\ 0 & \text{otherwise} \end{cases}$$

Proposition 1.3. Give an example to show that $j_p\mathcal{F}$ need not be a sheaf. Show that $j_!\mathcal{F}$ has stalks given by

$$(j_!\mathcal{F})_P = \begin{cases} \mathcal{F}_P & \text{if } P \in U \\ 0 & \text{otherwise} \end{cases}$$

Proof. We consider $X = \operatorname{Spec}(k[x, y]/(xy))$ and let $U = D(x)$ and $V = D(y)$, and $j : D(x) \hookrightarrow X$. In this case $U \cap V = \emptyset$, so if $j_p\mathcal{F}$ is a sheaf then its sections on U and V automatically glue to a section on $U \cup V$. But clearly $0 \neq x \in j_p\mathcal{F}(U) = k[x, y]_x/(xy)$ and $0 \in j_p\mathcal{F}(V)$ can't glue, because $j_p\mathcal{F}(U \cup V) = 0$.

For $P \in U$, $(j_!\mathcal{F})_P = (j_p\mathcal{F})_P = (j_p\mathcal{F}|_U)_P = \mathcal{F}_P$. For $P \notin U$, $j_p\mathcal{F}(W) = 0$ for any open neighborhood W of P . So $(j_!\mathcal{F})_P = (j_p\mathcal{F})_P = 0$ $\square$

Proposition 1.4. For sheaves of abelian groups $\mathcal{F}$ on U and $\mathcal{G}$ on X , construct a natural morphism $j_!j^{-1}\mathcal{G} \rightarrow \mathcal{G}$. Show that there is a natural isomorphism $\mathcal{F} \xrightarrow{\sim} j^{-1}j_!\mathcal{F}$

Proof. Note that for j an inclusion of an open set, $j^{-1}\mathcal{G}$ agrees with the restriction $\mathcal{G}|_U$. Now by universal property of sheafification, it suffices to give a morphism $j_pj^{-1}\mathcal{G} \rightarrow \mathcal{G}$. For $V \not\subset U$, $j_pj^{-1}\mathcal{G}(V) = 0$. For $V \subset U$, $j_pj^{-1}\mathcal{G}(V) = j^{-1}\mathcal{G}(V) = \mathcal{G}|_U(V) = \mathcal{G}(V)$. So we define $f : j_pj^{-1}\mathcal{G} \rightarrow \mathcal{G}$ by $f(V) : 0 \rightarrow \mathcal{G}(V)$ for $V \not\subset U$ and $f(V) : \mathcal{G}(V) \rightarrow \mathcal{G}(V)$ for $V \subset U$, $j_pj^{-1}\mathcal{G}(V) = \mathcal{G}(V) \xrightarrow{id} \mathcal{G}(V)$ for $V \subset U$

For $\mathcal{F} \rightarrow j^{-1}j_!\mathcal{F}$ and V open in U , note that $j^{-1}j_!\mathcal{F}(V) = j_!\mathcal{F}|_U(V) = j_!\mathcal{F}(V)$. And we have the natural morphism of sheafification $j_p\mathcal{F} \rightarrow j_!\mathcal{F}$. So we define $g : \mathcal{F} \rightarrow j^{-1}j_!\mathcal{F}$ by $g(V) : \mathcal{F}(V) =$

$j_p \mathcal{F}(V) \rightarrow j^{-1} j_! \mathcal{F}(V) = j_! \mathcal{F}(V)$. This is an isomorphism of sheaves because we have for each $P \in U$, $j^{-1} j_! \mathcal{F}_P = (j_! \mathcal{F}|_U)_P = j_! \mathcal{F}_P = \mathcal{F}_P$ by proposition 1.3, and so $g_P : \mathcal{F}_P \xrightarrow{\sim} (j^{-1} j_! \mathcal{F})_P$. So g is an isomorphism. $\square$

Proposition 1.5. *Let $Z = X \setminus U$ and $i : Z \hookrightarrow X$ be the inclusion. Let $\mathcal{G}$ be a sheaf of abelian groups on X . Show that there is an exact sequence of sheaves*

$$0 \rightarrow j_! j^{-1} \mathcal{G} \rightarrow \mathcal{G} \rightarrow i_* i^{-1} \mathcal{G} \rightarrow 0$$

Proof. Note that for $V \subset U$, $i_* i^{-1} \mathcal{G}(V) = i^{-1} \mathcal{G}(Z \cap V = \emptyset) = 0$ and for $V \not\subset U$, $j_! j^{-1} \mathcal{G}(V) = 0$. So this is a chain complex. Now it suffices to show that $(i_* i^{-1} \mathcal{G})_P = \mathcal{G}_P$ when $P \in Z$ and $(i_* i^{-1} \mathcal{G})_P = 0$ when $P \in U$, since by checking at the level of stalks we have

$$0 \rightarrow (j_! j^{-1} \mathcal{G})_P \xrightarrow{\sim} \mathcal{G}_P \rightarrow 0 \rightarrow 0$$

when $P \in U$ and

$$0 \rightarrow 0 \rightarrow \mathcal{G}_P \xrightarrow{\sim} (i_* i^{-1} \mathcal{G})_P \rightarrow 0$$

when $P \in Z$ exact in both cases.

Now for $P \notin Z$ and W an open neighborhood of P , $W \cap U$ is also an open neighborhood of P and $i_* i^{-1} \mathcal{G}(W \cap U) = 0$. So for $(W, f) \in (i_* i^{-1} \mathcal{G})_P$, $(W, f) \sim (W \cap U, f|_{W \cap U}) = 0$. So $(i_* i^{-1} \mathcal{G})_P = 0$.

For $P \in Z$, $(i_* i^{-1} \mathcal{G})_P = \varinjlim_{P \in W \text{ open}} i_* i^{-1} \mathcal{G}(W) = \varinjlim_{P \in W} i^{-1} \mathcal{G}(W \cap Z) = \varinjlim_{P \in W} \varinjlim_{W \cap Z \subset V} \mathcal{G}(V)$. Since $P \in W \cap Z \subset V$, we have $(i_* i^{-1} \mathcal{G})_P \subset \mathcal{G}_P$. On the other hand, for each open neighborhood W of P , we have $P \in W \cap Z \subset W$, i.e. $\mathcal{G}(W)$ appears in the first inductive limit. So both agree and the result follows. $\square$

Proposition 1.6. *Suppose further that X is a scheme, $U \subset X$ an open subscheme. Show that $j_! \mathcal{O}_U$ need not be a quasi-coherent sheaf on X .*

Proof. Consider $X = \mathbb{A}_k^1 = \text{Spec } k[x]$. Let $U = D(x)$. Suppose $j_! \mathcal{O}_U$ is quasi-coherent. Now, since X is affine, $j_! \mathcal{O}_U \cong \tilde{M}$ for some $k[x]$ -module M . Then $M = \Gamma(X, \tilde{M}) = j_! \mathcal{O}_U(X) = 0$. But $j_! \mathcal{O}_U(U) = \mathcal{O}_U(U) \cong k[x]_x \neq 0 = M_x = \Gamma(D(x), \tilde{M})$, contradiction. (Note that $j_p \mathcal{F}|_U$ is a sheaf so $j_p \mathcal{F}|_U = (j_p \mathcal{F}|_U)^+ = j_! \mathcal{F}|_U$. In particular we have $\mathcal{F}(U) = j_p \mathcal{F}(U) = j_! \mathcal{F}(U)$) $\square$

2 Schemes

Let A be a ring and $X = \text{Spec}(A)$. We know in Zariski topology X has an open basis $D(f)$ with $f \in A$. Define a sheaf $\mathcal{O}_X$ by letting $\mathcal{O}_X(U)$ be the set of locally quotient functions $s : U \rightarrow \bigsqcup_{p \in U} A_p$ with $s(p) \in A_p$. We call $\mathcal{O}_X$ the **structure sheaf** of X .

Some dirty work in commutative algebra tells us that $\mathcal{O}_{X,p} = A_p$ and $\mathcal{O}_X(D(f)) = A_f$. Particularly, we see the global sections of $\mathcal{O}_X$ is A .

A **ringed space** is a pair $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is a sheaf of rings on X . A **morphism of ringed space** $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is the data of a continuous map $f : X \rightarrow Y$ and a morphism of sheaves $f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$.

A **locally ringed space** is a ringed space $(X, \mathcal{O}_X)$ such that each stalk $\mathcal{O}_{X,x}$ is a local ring. A **morphism of locally ringed space** is a morphism of ringed space such that $f_x^\# : \mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$ is a local homomorphism, i.e. $f_x^{\#-1}(m_x) = m_y$ where $y = f(x)$. It induces an embedding of residue fields $k(y) \rightarrow k(x)$.

Example 2. Above discussions showed that $(\text{Spec} A, \mathcal{O}_{\text{Spec} A})$ is a locally ringed space, which is called the **affine scheme**. The category of affine schemes with locally ringed morphisms is equivalent to the opposite category of rings. The proof of fully-faithfulness consists of several tedious diagram chase.

Definition 2.1. A **scheme** is a locally ringed space $(X, \mathcal{O}_X)$ which is locally isomorphism to affine schemes, i.e. we can find an open covering $\{(U_i, \mathcal{O}_X|_{U_i})\}$ such that each $(U_i, \mathcal{O}_X|_{U_i})$ is isomorphic to an affine scheme.

Here we need to verify that $(U_i, \mathcal{O}_X|_{U_i})$ is a scheme. For $x \in U$, take $W \subset X$ an affine open nbhd of x . Then $U \cap W$ is open in W , so there exists a distinguished open set in $U \cap W \subset U$ containing x , which is affine

Example 3. A **k-point** on X is a morphism $\text{Spec}(k) \rightarrow X$. Let x be the image. Then the local homomorphism at stalks gives an inclusion $\kappa(x) \hookrightarrow k$. Conversely, an inclusion $\kappa(x) \hookrightarrow k$ also gives a k -point.

2.1 Gluing

I find myself always taking gluing procedure for granted after proving a theorem in affine. Although the gluing definitely works since the given theorems holds for any schemes, I feel like I never truly understand how it works, partially because of the puzzling wording in Hartshorne. Now I want to get over it.

Lemma 2.1. (*Gluing sheaves*) Let X be a topological space with an open cover $\{U_i\}$. Suppose we are given sheaves $\mathcal{F}_i$ on each U_i and sheaf isomorphisms $\phi_{ij} : \mathcal{F}_i|_{U_{ij}} \rightarrow \mathcal{F}_j|_{U_{ij}}$ where $U_{ij} = U_i \cap U_j$ satisfying the **cocycle condition**: $\phi_{ik} = \phi_{jk} \circ \phi_{ij}$ on $U_{ijk} = U_i \cap U_j \cap U_k$. Then there exists a sheaf $\mathcal{F}$ on X such that $\mathcal{F}|_{U_i} \cong \mathcal{F}_i$.

We can directly construct $\mathcal{F}$ by letting $\mathcal{F}(V) = \{(s_i)_i | s_i \in \mathcal{F}_i(V \cap U_i), \phi_{ij}(s_i|_{V \cap U_{ij}}) = s_j|_{V \cap U_{ij}}\}$ for V open in X . Now we give an isomorphism $\mathcal{F}_i \rightarrow \mathcal{F}|_{U_i}$: given V open in U_i and $s \in \mathcal{F}_i(V)$, define its image in $\mathcal{F}(V \cap U_i)$ to be $(\phi_{ij}(s|_{V \cap U_j}))$. This is well defined because $\phi_{jk} \circ \phi_{ij}(s|_{V \cap U_{jk}}) = \phi_{ik}(s|_{V \cap U_k})|_{V \cap U_{jk}}$ by cocycle condition.

Remark 2.1. The “cocycle” comes from 1-cocycle of group cohomology in the homogeneous definition.

Lemma 2.2. (*Gluing schemes*) Let $(X_i, \mathcal{O}_{X_i})$ be a collection of schemes. For each i, j suppose there exists open $U_{ij} \in X_i$ and an isomorphism of locally ringed space $\phi_{ij} : U_{ij} \rightarrow U_{ji}$. For each i, j, k , suppose $\phi_{ij}^{-1}(U_{jk} \cap U_{ji}) = U_{ij} \cap U_{ik}$ and $\phi_{ik} = \phi_{jk} \circ \phi_{ij}$ on $U_{ij} \cap U_{ik}$. Then there exists a scheme X with open covering (U_i) and isomorphisms of locally ringed space $\phi_i : X_i \rightarrow U_i$ such that $\phi_i(U_{ij}) = U_i \cap U_j$ and $\phi_{ij} = \phi_j^{-1} \phi_i$.

Indeed $X = (\bigsqcup X_i) / \sim$. For $x \in X_i, x' \in X_j, x \sim x'$ if and only if $x \in U_{ij}, x' \in U_{ji}$ and $\phi_{ij}(x) = x'$. So we are gluing (X_i) along U_{ij} 's. Moreover, we get $\mathcal{O}_{U_i}|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{O}_{U_j}|_{U_i \cap U_j}$ by identifying U_{ij} and U_{ji} and $U_i \cap U_j$. They also satisfy the cocycle condition so glue to the structure sheaf $\mathcal{O}_X$ on X .

To have a flavor on how this works, we start with the case when only two schemes $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are involved. Suppose U open in X, V open in Y and an isomorphism of locally ringed space $U \xrightarrow{\sim} V$. Then we can glue X, Y by identifying U and V . The cocycle condition is automatic since there is only two schemes.

Example 4. Given two affine lines $U_1 = \text{Spec}(k[x])$ and $U_2 = \text{Spec}(k[y])$, we glue them along $\mathbb{G}_m = \text{Spec}(k[t, t^{-1}])$ by $x \mapsto t$ and $y \mapsto t$. The resulting scheme X is the so-called the line with double origins. Then $\Gamma(X, \mathcal{O}_X) = k[x]$. Indeed, the s.e.s. of the sheaf condition on $\mathcal{O}_X$ give the global sections

are of the form (f, g) with $f \in k[x]$, $g \in k[y]$ such that $f|_{U_1 \cap U_2} = g|_{U_1 \cap U_2}$. But by our way of gluing, this implies that $f = g$ if x is identified with y .

If we glue along $\mathbb{G}_m$ by $x \mapsto t$ and $y \mapsto t^{-1}$, the resulting scheme is $\mathbb{P}_k^1$. Then for a global section (f, g) on $\mathbb{P}_k^1$, $f|_{U_1 \cap U_2}$ is a polynomial in t while $g|_{U_1 \cap U_2}$ is a polynomial in t^{-1} . So $f|_{U_1 \cap U_2} = g|_{U_1 \cap U_2}$ implies that they are both constant. So $f = g \in k$, and $\Gamma(\mathbb{P}_k^1, \mathcal{O}_{\mathbb{P}_k^1}) = k$.

Lemma 2.3. (*Gluing morphisms*) Given the gluing data in lemma 2.2, suppose we have morphisms $f_i : X_i \rightarrow Y$ such that $f_j \circ \phi_{ij} = f_i$, then there exists a morphism $f : X \rightarrow Y$ such that $f_i = f|_{U_i}$ by identifying X_i and U_i .

The moral is that a collection of morphisms can glue if they agree on the overlaps (identified parts), which is pretty intuitive.

Corollary 2.1. Let (U_i) be an open covering of X . Suppose we have $f_i : U_i \rightarrow Y$ with $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$. Then there exists a unique morphism $f : X \rightarrow Y$ such that $f|_{U_i} = f_i$.

In this case $U_{ij} = U_i \cap U_j = U_{ji}$ and ϕ_{ij} 's are identity map, so everything is clear.

We want to use this to prove several results, starting with

Proposition 2.1. The natural map $\text{Hom}_{\text{sch}}(X, \text{Spec} A) \rightarrow \text{Hom}_{\text{ring}}(A \rightarrow \Gamma(X, \mathcal{O}_X))$ is a bijection.

First we need a very important lemma on the intersection of affines open subschemes, so that we can glue

Lemma 2.4. Let X be a scheme, with open affine subsets $U = \text{Spec} A$, $V = \text{Spec} B$. Show that $U \cap V$ can be covered by open affine subschemes U_i such that there are elements $f_i \in A, g_i \in B$ with $U_i = D(f_i) \subset U$ and $U_i = D(g_i) \subset V$.

Proof of lemma. Let $x \in U \cap V$. Since $U \cap V$ open in U , there exists $f \in A$ such that $x \in D(f) \subset U \cap V$ and similarly $x \in D(g) \subset U \cap V$ for some $g \in B$. This induces an inclusion $i : D(f) \hookrightarrow V$ and hence a homomorphism $\varphi : B \rightarrow A_f$. So $D(f) \cap D(g) = i^{-1}D(g) = D(\varphi(g))$. Since $D(f)$ open in V , we assume that $D(g) \subset D(f)$. Then $D(g) = D(\varphi(g))$. Now $\varphi(g) = f'/f^n$ for some $f' \in A$ and n . Then $D(\varphi(g)) = D(f') \cap D(f)$. But $D(f') \cap D(f) = D(ff')$. So $D(g) = D(ff')$ (set theoretically) as desired. $\square$

Proof of proposition 2.1. We need to find an inverse to the natural map. Consider an affine open covering $\{U_i = \text{Spec} B_i\}$ of X . Given $\phi : A \rightarrow \Gamma(X, \mathcal{O}_X)$, by restricting to U_i we get homomorphisms $\phi_i : A \rightarrow B_i$, which induces $f_i : U_i \rightarrow \text{Spec}(A)$. We want to show f_i 's can glue, i.e. $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$.

By lemma 2.4, we can cover $U_i \cap U_j$ by open affine V such that $V = D(f) \subset U_i$ and $V = D(f') \subset U_j$. Then $f_i|_V$ is induced by $A \rightarrow B_{if}$ and $f_j|_V$ is induced by $A \rightarrow B_{jf'}$. But these two maps agree, since they are both the composition of ϕ and the restriction $\Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(V, \mathcal{O}_X)$. So f_i and f_j agree on V and hence on $U_i \cap U_j$. So they glue. $\square$

2.2 Projective Schemes

A **graded ring** S is $S = \bigoplus_{d \geq 0} S_d$ with S_d abelian groups such that $S_d \cdot S_{d'} \subset S_{d+d'}$. The graded rings are approximations to polynomial rings which have a natural grading structure.

A **homogeneous ideal** is an ideal I such that $I = \bigoplus_{d \geq 0} (I \cap S_d)$, i.e. I is generated by its homogeneous elements, and hence any element in I splits into homogeneous elements in I .

The **irrelevant ideal** $S_+ = \bigoplus_{d \geq 1} S_d$

Definition 2.2. Let $\text{Proj}(S)$ be the set of homogeneous prime ideals of S which do not contain S_+ . We equip $\text{Proj}(S)$ with the topology whose closed sets are of the form $V(I) = \{p \in \text{Proj}(S) \mid p \supset I\}$ where I is a homogeneous ideal

For $p \in \text{Proj}(S)$, let T be the set of homogeneous elements in $S \setminus p$. Then T is multiplicative, and let $S_{(p)} \subset T^{-1}S$ be the subring of homogeneous degree-0 elements. For $f \in S$ homogeneous, we write $S_{(f)} \subset S_f$ the subset of homogeneous degree-0 elements.

We define the structure sheaf on $\text{Proj}(S)$ as in affine. For U open, let the sections on U $\mathcal{O}(U)$ be the locally quotient maps $s : U \rightarrow \prod_{p \in U} S_{(p)}$ with $s(p) \in S_{(p)}$ and hence homogeneous of degree 0.

One can check we still have $\mathcal{O}_{X,p} = S_{(p)}$ and hence $(\text{Proj}(S), \mathcal{O})$ is a locally ringed space. Moreover, let $D_+(f) = \text{Proj}(S) \setminus V(f)$ for homogeneous f . Then we still have $(D_+(f), \mathcal{O}|_{D_+(f)}) \cong \text{Spec} S_{(f)}$. Since they form an open covering of $\text{Proj}(S)$, we see that $(\text{Proj}(S), \mathcal{O})$ is a scheme.

For a ring A , we define $\mathbb{P}_A^n = \text{Proj}(A[x_0, \dots, x_n])$.

2.3 Open and Closed Subschemes

An **open subscheme** of a scheme X is a scheme $(U, \mathcal{O}_X|_U)$ with U open in X . An **open immersion** is a morphism $X \rightarrow Y$ which induces an isomorphism of schemes of X and an open subscheme of Y .

Example 5. Open subscheme of affine scheme need not to be affine. A classic example is the “affine punctured plane”. Let k algebraically closed and $X = \mathbb{A}_k^2 = \text{Spec}(k[x, y])$. The origin O is a closed point and so $Z = X \setminus \{O\}$ is an open subscheme, which is the union of $D(x) \cong \text{Spec}(k[x, y]_x)$ (removing x -axis) and $D(y) \cong \text{Spec}(k[x, y]_y)$ (removing y -axis), with the intersection $D(x) \cap D(y) = D(xy) \cong \text{Spec}(k[x, y]_{xy})$. By the sheaf condition, Z 's global section is of the form (f, g) with $f \in k[x, y, y^{-1}]$ and $g \in k[x, x^{-1}, y]$ such that $f = g \in k[x, y, x^{-1}, y^{-1}]$. But this implies that $f = g \in k[x, y]$ and hence $\Gamma(Z, \mathcal{O}_Z) = k[x, y]$. So Z is not affine.

A **closed immersion** is a morphism $f : X \rightarrow Y$ if the induced topological map is a homeomorphism of X with a closed subset of Y , and $f^\#$ is surjective. A **closed subscheme** is an equivalence class of closed immersions, which we don't really care.

Remark 2.2. $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ surjective implies that $f_*\mathcal{O}_X \cong \mathcal{O}_Y/\mathcal{I}$ (we won't define quotient of sheaves) on $f(X)$ for some sheaf of ideals $\mathcal{I}$, which is called the **defining sheaf of ideals** of X

Theorem 2.1. ([Har77] II 3.11 (b)) A closed subscheme of an affine scheme $X = \text{Spec}(A)$ is affine. In fact, there is a one-to-one correspondence between ideals of A and closed subschemes of X , by $I \mapsto$ the image of $\text{Spec}(A/I)$ in X with the natural closed immersion, i.e. any closed subscheme of $\text{Spec}(A)$ is isomorphic to some $\text{Spec}(A/I)$

Hartshorne gives two approaches to theorem 2.1. One is to use the criterion of affineness in exercise 2.17, where he defines an open set X_f consisting of points where f doesn't vanish for a global section f (the **support** of f), which is the generalization of $D(f)$ to schemes. The other uses an important result on quasi-coherent sheaves, which says that locally they are the associated sheaves of modules. In our case for a closed subscheme over $\text{Spec}(A)$ with defining sheaf of ideals $\mathcal{I}$, $\mathcal{I}$ is the associated sheaf of an ideal I . Then $\mathcal{O}_X/\mathcal{I} \cong \mathcal{O}_{\text{Spec}(A/I)}$. We can check the support of $\mathcal{O}_X/\mathcal{I}$ to see that the image is $V(I)$. So we get an inverse to the map in the statement.

2.4 Fibre Product

Let $X \rightarrow S, Y \rightarrow S$ be morphisms in some category $\mathcal{C}$. The **fibre product** of them is an object $X \times_S Y$ with projections p_X and p_Y to make the following diagram commute

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{p_X} & X \\ \downarrow p_Y & & \downarrow \\ Y & \longrightarrow & S \end{array}$$

with the universal property that for an object Z with commutative diagram

$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \longrightarrow & S \end{array}$$

there exists

a unique morphism $Z \rightarrow X \times_S Y$ to make the bigger diagram commute, i.e. the fibre product is the final object in the category of objects with this property.

Theorem 2.2. *Fibre products always exist in the category of schemes.*

The idea of the proof is to prove the case in affine and glue, which is more or less a standard procedure for these theorems, although we need to note that for the fibre product of $\text{Spec} A$ and $\text{Spec}(B)$ over $\text{Spec}(R)$ with A, B R -algebra is precisely $\text{Spec}(A \otimes_R B)$.

Example 6. Consider $\mathbb{P}_k^1 \times \mathbb{A}_k^1 = \text{Proj}(k[x_0, x_1]) \times_k \text{Spec}(k[x])$. $\mathbb{P}_k^1$ is covered by two affine charts $\text{Spec}(S_{(x_0)}) = \text{Spec}(k[x_1/x_0])$ and $\text{Spec}(k[x_1/x_0])$ and their fibre products with $\text{Spec}(k[x])$ is $\text{Spec}(k[x][x_0/x_1])$ and $\text{Spec}(k[x][x_1/x_0])$. They glue to get $\mathbb{P}_k^1 \times \mathbb{A}_k^1 = \text{Proj}(k[x][x_0, x_1])$, where x is of degree 0 and x_0, x_1 are of degree 1.

In general, $\text{Proj}(S) \times_k \text{Spec}(A) = \text{Proj}(S \otimes_k A)$ where S is a k -graded algebra with $S_0 = k$. We can think of this as base change which makes A the new S_0 and $S \otimes_k A$ then has a natural grading structure from S .

We also note that $\text{Proj}(S_1) \times_k \text{Proj}(S_2) = \text{Proj}(S_1 \hat{\otimes}_k S_2)$ with $S_1 \hat{\otimes}_k S_2$ the bigraded ring, which is analogous to the construction of total complex in homological algebra.

Given an S -scheme X and a morphism $S' \rightarrow S$, $X \times_S S'$ is an S' -scheme and this process is called **base change**. Many properties are stable after base change (see [Poo17, Appendix C]). Moreover an S -morphism $X \rightarrow Y$ naturally induces an S' -morphism $X \times_S S' \rightarrow Y \times_S S'$

Proposition 2.2. *Closed immersions are stable after base change.*

This is more difficult than it looks like (in fact, most proofs of showing some property is stable after base change are annoying). Because of theorem 2.1, locally we can work with everything affine and this again is reduced to commutative algebra.

For a morphism $f : X \rightarrow Y$, let $y \in Y$ with residue field $\kappa(y)$. We can define a morphism $\kappa(y) \rightarrow Y$ with image y . The fibre product $X_y = X \times_Y \text{Spec}(\kappa(y))$ is called the (scheme-theoretically) fibre of f at y .

Lemma 2.5. X_y is homeomorphic to $f^{-1}(y)$.

To prove this we need another lemma

Lemma 2.6. *Let U open in X . Then the fibre product $U \times_S Y = p_X^{-1}(U)$*

The universal property of $p_X^{-1}(U)$ can be seen from the diagram

$$\begin{array}{ccccc} T & \xrightarrow{\phi} & X \times_S Y & \xrightarrow{p_Y} & Y \\ \downarrow & & \downarrow p_X & & \downarrow \\ U & \hookrightarrow & X & \longrightarrow & S \end{array}$$

and we see that T lands in $p_X^{-1}(U)$ through ϕ .

Example 7. Consider $f : X = \text{Spec}(k[t]) \rightarrow Y = \text{Spec}(k[x])$ induced by $k[x] \rightarrow k[t]$, $x \mapsto t^2$. Let $y = \langle x - a \rangle$. Then $\kappa(y) = k[x]/\langle x - a \rangle \cong k$ and $X_y = \text{Spec}(k[x]/\langle x - a \rangle \otimes_{k[x]} k[t]) = \text{Spec}(k[t]/\langle t^2 - a \rangle)$

Note that we used the following lemma which is helpful when computing the fibres or fibre product

Lemma 2.7. *Let A, B two rings and B an A -algebra with the structure map $\phi : A \rightarrow B$. Then for any ideal $I \subset A$ we have*

$$A/I \otimes_A B \cong B/IB$$

Here $IB = \phi(I) \cdot B$

2.5 Exercise

In the following we compute the fibres of different morphisms. Here k is assumed to be algebraically closed.

1. Consider $\text{Spec}(k[T, V]/(TV - 1)) \rightarrow \text{Spec}(k[T])$ induced by the natural map. For a maximal ideal $x = (T - a) \subset k[T]$, so $\kappa(x) \cong k[T]/(T - a)$, and so the fibre over x is $\text{Spec}(k[T, V]/(TV - 1)) \otimes_{k[T]} k[T]/(T - a) \cong k[V]/(aV - 1)$. For $x = (0)$, $\kappa(x) = k(T)$, and the special fibre is $\text{Spec}(k(T)[V]/(TV - 1))$
2. Let $X = \mathbb{P}_{\mathbb{F}_p}^2$ and $F : X \rightarrow X$ be the Frobenius, i.e. identity on the topological space and take the p -th power on the ring of functions. We pick $x = (x_1/x_0) \in D_+(x_0)$. Then $\kappa(x) = \mathbb{F}_p[x_1/x_0]/(x_1/x_0)$. We consider the fibre over x . Since the set-theoretical fibre of x is just $x \in D_+(x_0)$, we can restrict to $D_+(x_0)$ and compute $D_+(x_0) \times_F \text{Spec}(\kappa(x)) = \text{Spec}(\mathbb{F}_p[x_1/x_0] \otimes_{\mathbb{F}_p[x_1/x_0], F} \mathbb{F}_p[x_1/x_0]/(x_1/x_0)) \cong \text{Spec}(\mathbb{F}_p[x_1/x_0]/(x_1/x_0)^p)$. We also note that the global sections on X is just $\mathbb{F}_p$, so the Frobenius on them is just identity.

3 Properties of Schemes

A scheme X is **integral** (resp. **reduced**) if for each U open in X $\mathcal{O}_X(U)$ is an integral domain (resp. reduced). It is **irreducible** if its underlying topological space is irreducible, i.e. it cannot be written as the union of two proper closed sets.

Theorem 3.1. *A scheme X is integral if and only if it is reduced and irreducible.*

Proof. If X is integral, it is clear that it is reduced. Suppose it contains two disjoint open U_1, U_2 , then sheaf conditions give $\mathcal{O}_X(U_1 \cup U_2) = \mathcal{O}_X(U_1) \times \mathcal{O}_X(U_2)$, which is not an integral domain.

Conversely, suppose X is reduced and irreducible. For any open $U \subset X$, suppose that $fg = 0$ for $f, g \in \mathcal{O}_X(U)$. Define $Y = U \setminus X_f$ where f vanishes inside U , and Z where g vanishes inside U , which are closed in U . Then fg vanishes on U implies that $Y \cup Z = U$. Note that U is irreducible since U is open in X which is irreducible. So we have $Y = U$ or $Z = U$. WLOG assume that $Y = U$. Then for any affine open $V = \text{Spec}(A) \subset U$, $f|_V$ vanishes on V implies that $f|_V$ is in the nilradical of A and hence nilpotent and hence $f|_V = 0$ since X is reduced. Hence $f = 0$. $\square$

In the proof we see that being reduced is a local property, i.e. X is reduced if and only if every open affine $\text{Spec}(A) \subset X$ has A reduced if and only if every local ring is reduced. On the other hand, being irreducible is a topological property. So being integral is not a local property, and local rings are integral domain is not enough for a scheme to be integral.

Lemma 3.1. *A is an integral domain iff $X = \text{Spec}(A)$ is integral*

Proof. If A is a domain, suppose $V(I) \cap V(J) = V(IJ) = X$. Then $IJ \subset \text{nil}(A) = 0$ and hence $I = 0$ or $J = 0$ since A is a domain. So X is irreducible, and reduced by checking stalks, and so is integral. The other direction is clear. $\square$

Theorem 3.2. *For an integral scheme X , there exists a unique $\eta \in X$ such that $\overline{\{\eta\}} = X$. This is called the **generic point** of X . The stalk at η is a field called the **function field** of X , denoted by $K(X)$.*

Proof. Take any affine open $U = \text{Spec}(A) \subset X$. Then A is an integral domain and the point (0) is a generic point of U . Its stalk is $A_{(0)}$ and is the fractional field of A . Let (0) be η in X . First note that $U \subset \overline{\{\eta\}}$. Then $X = \overline{\{\eta\}} \cup (X \setminus U)$. Since X is irreducible and U is nonempty, this implies that $\overline{\{\eta\}} = X$ and hence η is in every open subset of X . Suppose η' is another point such that $\overline{\{\eta'\}} = X$, then η' is in every open subset and, in particular, U . Also η' is the generic point of U . But $U = \text{Spec} A$ only has a unique generic point $(0) = \eta$. So $\eta = \eta'$ $\square$

Remark 3.1. On the other hand, if X has a generic point, then clearly X is irreducible.

Proposition 3.1. *If S is a graded domain and $S_+ \neq 0$, then $\text{Proj}(S)$ is integral*

Proof. By assumption (0) is a homogeneous prime ideal, and so is the generic point of $\text{Proj} S$. Hence $\text{Proj} S$ is irreducible. $\text{Proj} S$ can be covered by $D_+(f) \cong \text{Spec}(S_{(f)})$ with f a homogeneous element. Then $S_{(f)}$ is integral and hence reduced and so $D_+(f)$ is reduced. Since being reduced is a local property, $\text{Proj} S$ is reduced, and so integral $\square$

Remark 3.2. It is not true that S is necessarily a domain if $\text{Proj} S$ is integral. Indeed, for S domain we have $\text{Proj} S = \text{Proj} S[x]/x^2$. See [Har77, II, Exercise 5.14]

A scheme is **locally Noetherian** if it can be covered by open affine $U_i = \text{Spec}(A_i)$ with A_i Noetherian. It is **Noetherian** if it is also quasi-compact, i.e. it can be covered by finite many open affine $U_i = \text{Spec}(A_i)$ with A_i Noetherian.

Being locally noetherian is a local property. A scheme X is locally noetherian if and only if for every affine open $U = \text{Spec}(A) \subset X$, A is noetherian.

A morphism $f : X \rightarrow Y$ is **locally of finite type** if there exists a covering of Y by open affine subsets $V_i = \text{Spec}(B_i)$ such that each $f^{-1}(V_i)$ can be covered by affine open $U_{ij} = \text{Spec}(A_{ij})$ with each A_{ij} finitely generated B_i -algebra. f is **of finite type** if for each i , the cover of $f^{-1}(V_i)$ above can be taken to be finite.

Being locally of finite type is a local property. $f : X \rightarrow Y$ is locally of finite type if and only if for every affine open $V = \text{Spec}(B) \subset Y$, $f^{-1}(V)$ can be covered by affine open $U_i = \text{Spec}(A_i)$ with A_i finitely generated B -algebra.

A **variety** is an integral scheme of finite type over $\text{Spec} k$, for k an algebraically closed field.

Remark 3.3. For A noetherian and B finitely generated A -algebra, B is a quotient by finitely many polynomials. Hence a scheme locally of finite type over R is the cutting out of finitely many polynomials, i.e. it is determined by finite data. This is where the name "finite type" comes from. If A is not noetherian, then we need **locally of finite presentations**, i.e. it is presented by finite data. This is explained in [Poo17] along with the naive "clearing denominator" method.

Example 8. We give two examples of a morphism $X \rightarrow Y$ which is locally of finite type but not of finite type. The simple one is to take X to be a disjoint union of a non-finite number of copies of $\text{Spec}(k)$ and $Y = \text{Spec}(k)$. Now we take $X_i = \text{Spec}k[x_i, y_i]/(x_i y_i)$. Let $X_i \supset U_{i,i-1} = D(x_i) \cong \text{Spec}(k[x_i, 1/x_i])$, and $X_i \supset U_{i,i+1} = D(y_i) \cong \text{Spec}k[y_i, 1/y_i]$. We give an identification $U_{i,i+1} \xrightarrow{\sim} U_{i+1,i}$ via the k -algebra isomorphism given by $y_i \mapsto x_{i+1}^{-1}$. This gives a gluing of schemes inducing an infinite chain of $\mathbb{P}^1$'s.

3.1 Separatedness and Properness

Recall that a topological space X is Hausdorff if and only if the diagonal $\Delta = \{x, x \mid x \in X\} \subset X \times X$ is closed. Separatedness is an analogy for Hausdorff for schemes.

A morphism $X \rightarrow S$ is **separated** if the diagonal map $\Delta_{X/S} : X \rightarrow X \times_S X$ is a closed immersion. We say an S -scheme X is separated if the structure morphism is separated.

If $X \rightarrow S$ is a morphism of affine schemes induced by $R \rightarrow A$, then by looking at the diagram we can see that the diagonal map is induced by $A \otimes_R A \rightarrow A$, $a \otimes b \mapsto ab$ (the unique map to make this commute).

Corollary 3.1. *Morphisms of affine schemes are separated*

Proof. The homomorphism $A \otimes_R A \rightarrow A$, $a \otimes b \mapsto ab$ is surjective, so $A \cong (A \otimes_R A)/I$ for some ideal I . $\square$

Corollary 3.2. *A morphism $f : X \rightarrow S$ is separated if and only if the image of the diagonal map $\Delta(X)$ is closed.*

Proof. Suppose $\Delta(X)$ is closed. We have $p_1 \circ \Delta = \text{id}_X$ since the diagram commutes, and hence X is homeomorphic to $\Delta(X)$. So it suffices to show that the sheaf morphism $\mathcal{O}_{X \times_S X} \rightarrow \Delta_* \mathcal{O}_X$ is surjective which is a local problem. So for $x \in X$, we can take an affine open neighborhood U of x small enough such that $f(U)$ contained in some open affine V in S . Now by corollary 3.1, $U \rightarrow V$ is separated and hence $\mathcal{O}_{U \times_V U} \rightarrow \Delta_* \mathcal{O}_U$ is surjective. So our map is surjective in a neighborhood of x , and we are done. $\square$

Remark 3.4. Open and closed immersions are separated. So are compositions of separated morphisms.

Proposition 3.2. *Let X be separated over $S = \text{Spec}A$ and U, V affine open. Then $U \cap V$ is affine.*

Proof. Let U, V be two affine open. The diagonal map Δ restricts to $\Delta : U \cap V \rightarrow U \times_S V$. This gives a homeomorphism of $U \cap V$ with $\Delta(X) \cap U \times_S V$ which is closed in $U \times_S V$ since X is separated, and the stalk maps are still surjective. So this is in fact a closed immersion. Since any closed subscheme of affine schemes is affine, we get the required result $\square$

Conversely, if we can find an affine cover $\{U_i\}$ of X such that each intersection $U_i \cap U_j$ is affine, and $\Gamma(U_i, \mathcal{O}_X)$ and $\Gamma(U_j, \mathcal{O}_X)$ generate $\Gamma(U_i \cap U_j, \mathcal{O}_X)$ as rings, then X is separated. So $\text{Proj} S$ is separated for any graded ring S .

Example 9. Consider the **affine plane with two origins**, i.e. two affine planes gluing along $\mathbb{A}_k^2 \setminus (0, 0)$. We have seen that the punctured plane is not affine, while it is the intersection of those two affine planes. So "affine plane with two origins" is not separated, as expected.

A morphism is **proper** if it is separated, of finite type and universally closed, i.e. closed after base change.

Example 10. In practice it is even not easy to tell a map is closed. For instance, we consider $p_x : \mathbb{A}_k^2 \rightarrow \mathbb{A}_k^1$ the projection to the x -axis. Then $p_x(V(xy - 1)) = D(x)$, which is clear from the picture, and so p_x is not closed.

From this, we see that the structure map $\mathbb{A}_k^1 \rightarrow \text{Spec}(k)$ is closed but not universally closed, because after base change to $\mathbb{A}_k^1$ this is exactly the projection $\mathbb{A}_k^2 \rightarrow \mathbb{A}_k^1$.

However, there is a useful criterion for separatedness, universally closedness and properness related to algebraic number theory

Theorem 3.3. (*Valuative Criterion*) *For $f : X \rightarrow Y$ of finite type (we don't need this for separatedness and universally closedness) with X noetherian, Then f is proper (resp. separated, universally closed) if and only if for any DVR R with fraction field K and a commutative diagram*

$$\begin{array}{ccc} \text{Spec}(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow f \\ \text{Spec}(R) & \longrightarrow & Y \end{array}$$

there exists a unique (resp. at most one, at least one) morphism $\text{Spec}(R) \rightarrow X$ that fits into the diagram.

Example 11. A line with double origins is not separated. This can be seen by taking $R = k[[t]]$ and $K = k((t))$. Then the generic point of the double-origin line is a K -point by $x \mapsto t$, which can extend to two morphisms $\text{Spec}(R) \rightarrow X$, namely mapping the special point to those two origins, since their residue fields are both k .

Corollary 3.3. *Closed immersions are proper.*

We have seen it is separated. It is universally closed because it is closed (homeomorphic to a closed set) and stable after base change.

Lemma 3.2. *Properness is stable after base change.*

Example 12. The projective space $\mathbb{P}_A^n$ is proper over A . We have seen it is separated. Since it can be covered by its natural affine covers, it is also of finite type. The hard part is to show it is universally closed. We consider $\mathbb{P}_{\mathbb{Z}}^n$. This is because $\mathbb{Z}$ is in any rings so would not cause any trouble, and base change won't affect properness. For $\mathbb{P}_{\mathbb{Z}}^1$, given a K -point, say it lands in $D_+(x_0)$. If the pullback of x_1/x_0 is 0, this is already an R -point. Otherwise, we see it actually lands in $D_+(x_0x_1)$, so if the pullback of x_0/x_1 has negative valuation, we can consider to pullback x_1/x_0 , so that this gives a R -point on $D_+(x_0)$ and hence on $\mathbb{P}_{\mathbb{Z}}^1$. In general, we can proceed by induction.

We can show $\mathbb{P}_A^n \rightarrow \text{Spec} A$ is closed instead, which does not seem any easier.

4 Sheaves of Modules

For a ringed space $(X, \mathcal{O}_X)$, a **sheaf of $\mathcal{O}_X$ -module** is a sheaf of abelian groups $\mathcal{F}$ on X such that for each open U is an $\mathcal{O}_X(U)$ -module with functoriality.

We use $\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ to denote the group of $\mathcal{O}_X$ -homomorphisms $\mathcal{F} \rightarrow \mathcal{G}$. We define the **sheaf hom** $\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ via $U \mapsto \text{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U)$. We can also define the tensor product of $\mathcal{O}_X$ -modules.

We have seen that for a morphism $f : X \rightarrow Y$, $f_*\mathcal{F}$ is naturally a $\mathcal{O}_Y$ -module, and how to define the pullback $f^*\mathcal{G}$.

For an A -module M , we can define its **associated sheaf** $\tilde{M}$ on $\text{Spec}(A)$ as how we define the structure sheaf. Then analogously we have $\tilde{M}(D(f)) = M_f$ and $(\tilde{M})_{p_x} = M_p$ and $\Gamma(X, \tilde{M}) = M$

A sheaf of $\mathcal{O}_X$ -module $\mathcal{F}$ is **quasi-coherent** if X can be covered by affine open $U_i = \text{Spec}(A_i)$ such that $\mathcal{F}|_{U_i} \cong \tilde{M}_i$ for some A_i -module M_i . It is **coherent** M_i 's can be taken finitely generated.

Remark 4.1. Quasi-coherent and coherent sheaves are analogy of modules and finitely generated modules.

Here is an important local property of quasi-coherent sheaves

Theorem 4.1. *Let $\mathcal{F}$ be an $\mathcal{O}_X$ -module. Then $\mathcal{F}$ is quasi-coherent if and only if for any affine open $U = \text{Spec}(A)$ in X , $\mathcal{F}|_U \cong \tilde{M}$ for some A -module M .*

So quasi-coherent sheaves are locally associated sheaves of module. Similar results hold for coherent sheaves and finitely generated modules when X is noetherian.

Moreover, for $X = \text{Spec}(A)$ we have an equivalence of categories between A -modules and quasicoherent sheaves on X given by $M \mapsto \tilde{M}$ with inverse $\mathcal{F} \mapsto \Gamma(X, \mathcal{F})$. In particular, we get a bijection $\text{Hom}_A(M, N)$ and $\text{Hom}_{\mathcal{O}_X}(\tilde{M}, \tilde{N})$. Since being quasicoherent is a local property, we can always reduce to the affine case, where it is easy to show that the kernel, cokernel and image of morphisms of quasicoherent sheaves are quasicoherent. More precisely, given $f : M \rightarrow N$ inducing $\tilde{f} : \tilde{M} \rightarrow \tilde{N}$, we have $\ker \tilde{f} = \tilde{\ker(f)}$, $\text{im} \tilde{f} = \tilde{\text{im} f}$ and $\text{coker} \tilde{f} = \tilde{\text{coker}(f)}$, which can be seen by checking at stalks.

Corollary 4.1. *For a s.e.s of quasis-coherent sheaves $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$, $0 \rightarrow \Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'') \rightarrow 0$ is exact*

A sheaf $\mathcal{F}$ is **locally free of order** r if there exists an open covering $\{U_i\}$ such that $\mathcal{F}|_{U_i} \cong \mathcal{O}_{U_i}^{\otimes r}$. They are also known as vector bundles. The cover is called the **trivialising cover**

A locally free sheaf of order 1 is called a **line bundle** or **invertible sheaf**

Example 13. The most important example of line bundle is $\mathcal{O}_{\mathbb{P}^n_k}(1)$. In general, for $X = \text{Proj}(S)$, we define $S(m)$ be the graded S -module with shifted degree by m , i.e. $(S(m))_d = S_{d+m}$. Then $\mathcal{O}_X(m) = S(m)$

We now compute $\Gamma(\mathbb{P}^n_k, \mathcal{O}_{\mathbb{P}^n_k}(1))$. Let $U = D_+(x_0)$ and $V = D_+(x_1)$. Then $\Gamma(U, \mathcal{O}_{\mathbb{P}^1}(1)) = k[x_0, x_1]_{x_0}(1)$ (degree-1 homogeneous elements), whose generators are $x_0, x_1, x_1^2/x_0, \dots$, and similarly for U' . Glue and we see the global sections of $\mathcal{O}_{\mathbb{P}^1}(1)$ consist of linear homogeneous polynomials in x_0, x_1 . More generally, $\Gamma(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m))$ consists of homogeneous degree- m polynomials in $x_0, x_1, \dots, x_n$

There is an equivalent definition by $\mathcal{O}_{\mathbb{P}^n_k}(m) = \mathcal{O}_{\mathbb{P}^n_k}(mH)$, where H is the hypersurface $V(x_0)$, which we are about to see.

For $\mathcal{L}$ a line bundle with trivalising cover $\{U_i\}$, on $U_i \cap U_j$ we have the transition map $\phi_{ij} : \mathcal{O}_{U_i}|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{L}|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{O}_{U_j}|_{U_i \cap U_j}$. Then ϕ_{ij} is an automorphsim of $\mathcal{O}_{U_i \cap U_j}$, which is given by multiplication by some $g_{ij} \in \mathcal{O}_X^*(U_i \cap U_j)$ as $\text{Hom}_A(A, A) = A$ where $\mathcal{O}_X^*$ is the sheaf of invertible sections of $\mathcal{O}_X$. It's clear that they satisfy the cocycle conditions, i.e. $\phi_{ik} = \phi_{jk} \circ \phi_{ij}$ and $g_{ik} = g_{jk}g_{ij}$. Conversely a set $\{g_{ij} \in \mathcal{O}_{U_i \cap U_j}^* \mid i, j\}$ satisfying the cocycle condition (i.e. a Čech 1-cocycle of $(\mathcal{U}, \mathcal{O}_X^*)$) determines a line bundle over $\{U_i\}$ up to isomorphism, and they give isomorphic line bundles iff they differ by a 1-coboundary. So we showed that $\text{Pic}(\mathcal{U}) = \check{H}^1(\mathcal{U}, \mathcal{O}_X^*)$ where $\text{Pic}(\mathcal{U})$ denotes the line bundles with trivializing cover $\mathcal{U}$ up to isomorphism.

Let $f : X \rightarrow Y$ be a morphism and $\mathcal{L}$ a line bundle on Y . It turns out the pullback $f^*\mathcal{L}$ is a line bundle. Indeed, for a trivalising cover $\{U_i\}$, let $V_i = f^{-1}(U_i)$ which gives a covering of X . Then $f^*(\mathcal{L})|_{V_i} = f^*(\mathcal{L}|_{U_i}) \cong f^*(\mathcal{O}_{U_i}) = f^{-1}\mathcal{O}_{U_i} \otimes_{f^{-1}\mathcal{O}_{U_i}} \mathcal{O}_{V_i} = \mathcal{O}_{V_i}$. So $\{V_i\}$ is a trivalising cover for $f^*\mathcal{L}$. The transition maps are given by $f^\#(g_{ij})$.

Remark 4.2. The pushforward of line bundles doesn't behave well. This follows from the fact that the pushforward of coherent sheaves need not to be coherent sheaves, even if they are always quasi-coherent when X is noetherian.

For two line bundles $\mathcal{L}_1, \mathcal{L}_2$ with a common trivialising cover, $\mathcal{L}_1 \otimes_{\mathcal{O}_X} \mathcal{L}_2$ is a line bundle with transition map $g_{ij}h_{ij}$. We can define the inverse $\mathcal{L}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$, which is a line bundle with transition map g_{ij}^{-1} . Then note that $\mathcal{L} \otimes \mathcal{L}^\vee$ is a line bundle with transition map 1, and hence $\mathcal{L} \otimes \mathcal{L}^\vee \cong \mathcal{O}_X$.

Therefore, we define the **Picard group** of X to be the isomorphism of line bundles. This is a group with group law given by tensor product and inverse given by taking the dual.

Remark 4.3. We can define $\mathcal{O}_{\mathbb{P}^n}(-1) = \mathcal{O}_{\mathbb{P}^n}(1)^\vee$. Note that $\Gamma(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(-1)) = 0$ by gluing sections.

We have also defined the **basepoint-free** line bundle or line bundles **globally generated** by $s_0, \dots, s_n$, i.e. base locus empty, in toric varieties notes. We have seen this induces a morphism $X \rightarrow \mathbb{P}^n$ by $x \mapsto [s_0(x) : \dots : s_n(x)]$. If this is a closed immersion, the line bundle $\mathcal{L}$ is called **very ample** and is the pullback of $\mathcal{O}_{\mathbb{P}^n}(1)$ through the immersion. It is **ample** if $\mathcal{L}^{\otimes k}$ is very ample for some k .

4.1 Exercise

Proposition 4.1. For $i : X \rightarrow Y$ a morphism with X noetherian and $\mathcal{F}$ a quasi-coherent sheaf on X , the pushforward $i_*\mathcal{F}$ is quasi-coherent

We need the following lemma to deal with the affine case

Lemma 4.1. Let $\phi : A \rightarrow B$ a ring homomorphism inducing $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$. For M a B -module, denote M_A the induced A -module with underlying group M , i.e. $a \cdot m = \phi(a)m$. Then $f_*(\tilde{M}) = \tilde{M}_A$

Proof. Take any distinguished open set $D(a) \subset \text{Spec}(A)$. $f_*(\tilde{M})(D(a)) = \tilde{M}(f^{-1}D(a)) = \tilde{M}(D(\phi(a))) \cong M_{\phi(a)}$. Also $\tilde{M}_A(D(a)) \cong M_{A,a}$, and we have an isomorphism $M_{\phi(a)} \rightarrow M_{A,a}; m/\phi(a)^n \mapsto m/a^n$. So they agree on distinguished open sets and hence are the same. $\square$

Remark 4.4. $f^*(\tilde{M}) = M \otimes_A B$

Proof of proposition 4.1. Since being quasi-coherent is a local property, we may assume $Y = \text{Spec}(A)$ is affine. Since X is noetherian, we can cover a finite affine cover $\{U_i = \text{Spec}(B_i)\}_{i \in I}$ of X such that $\mathcal{F}|_{U_i} \cong \tilde{M}_i$ with M_i a B_i -module. On the overlaps $U_i \cap U_j$ we can take a finite affine cover $\{U_{ijk}\}_{k \in I_{ij}}$ with $U_{ijk} = D(f_{ijk}) \subset U_i$ for $f_{ijk} \in B_i$

Now note that the morphism of sheaves $\oplus_{i \in I} \mathcal{F}|_{U_i} \rightarrow \oplus_{i,j \in I, k \in I_{ij}} \mathcal{F}|_{U_{ijk}}$ given by $(s_i) \mapsto (s_i|_{U_{ijk}} - s_j|_{U_{ijk}})$ has kernel $\mathcal{F}$. So we have an exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow \oplus_{i \in I} \mathcal{F}|_{U_i} \cong \tilde{M}_i \rightarrow \oplus_{i,j \in I, k \in I_{ij}} \mathcal{F}|_{U_{ijk}} \cong \tilde{M}_{i f_{ijk}}$$

Applying $f_*(-)$ which is left exact, we have an exact sequence

$$0 \rightarrow f_*\mathcal{F} \rightarrow \oplus_{i \in I} f_*(\tilde{M}_i) \rightarrow \oplus_{i,j \in I, k \in I_{ij}} f_*(\tilde{M}_{i f_{ijk}})$$

By the lemma, the last two terms are quasicohherent. So $f_*\mathcal{F}$ as the kernel of morphism of quasi-coherent sheaves is also quasi-coherent. $\square$

Proposition 4.2. *Let $\phi : \mathbb{P}^n \rightarrow \mathbb{P}^m$ be a morphism. Then either $\phi(\mathbb{P}^n)$ is a point or $m \geq n$ and $\dim(\phi(\mathbb{P}^n)) = n$*

Proof. Every morphism $\phi : \mathbb{P}^n \rightarrow \mathbb{P}^m$ is induced by a line bundle $\mathcal{L}$ globally generated by $m+1$ sections $s_0, \dots, s_m$ by Hartshorne II 7.1. Also line bundles on $\mathbb{P}^n$ are of the form $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n}(r)$. Since the global sections of $\mathcal{O}_{\mathbb{P}^n}(s)$ is 0 for $s < 0$, we have $r \geq 0$. Note that the sections s_i correspond to homogeneous polynomials f_i of degree r , and the common vanishing locus of all f_i is $V(f_0, \dots, f_m)$. But this is non-empty if $m+1 \leq n$ by dimension theory. Since $s_0, \dots, s_m$ has empty base locus, we have $m \geq n$.

If $\dim(\phi(\mathbb{P}^n)) < n \leq m$, we can pick $x \in \mathbb{P}^m \setminus \phi(\mathbb{P}^n)$ which induces a projection $\pi : \mathbb{P}^m \rightarrow \mathbb{P}^{m-1}$. Then we get a morphism $\pi \circ \phi : \mathbb{P}^n \rightarrow \mathbb{P}^{m-1}$. So we can reduce m till $m = n-1$, which then contradicts the first part. $\square$

5 Divisors

5.1 Wéil Divisors

For a topological space X , the **dimension** of X is the length of the longest chain of irreducible closed subsets of X . For example, $\dim \mathbb{A}^n = \dim \mathbb{P}^n = n$.

Given an irreducible closed subset $Z \subset X$, the **codimension** of Z in X is the length of the longest chain of closed irreducible subsets starting with Z .

Any closed irreducible subset admits a unique generic point, since irreducible closed sets in affine are of the form $V(\mathfrak{p})$. We can think of the codimension of a closed irreducible set as the height of its generic point. For $\eta \in Z$ its generic point, we have $\dim \mathcal{O}_{X,\eta} = \text{codim}(Z, X)$.

If X is any Noetherian topological space, then any closed subset $Z \subset X$ has a finite decomposition into irreducible components.

Now we make an assumption as in Hartshorne in this section: X is a Noetherian integral separated scheme which is regular in codimension 1. This means that for any local ring $\mathcal{O}_{X,x}$ with dimension 1, it is regular, i.e. $\dim_{\kappa(x)} m_x / m_x^2 = \dim \mathcal{O}_{X,x}$. This makes it a DVR. So this condition ensures that every codim-1 irreducible closed subset gives a discrete valuation on $K(X)$, which motivates the following definition.

A **prime divisor** on X is an integral closed subscheme of X of codimension 1. A **Weil divisor** on X is an element of the free abelian group $\text{Div} X$ on the set of prime divisors. It is a finite sum $\sum n_D D$ and **effective** if $n_D \geq 0$.

For any $f \in K(X)^\times$, we define its divisor $\text{div}(f) = \sum v_D(f) D$, where v_D is the discrete valuation of the DVR $\mathcal{O}_{X,\eta_D}$.

Proposition 5.1. *$\text{div}(f)$ is a Weil divisor, i.e. a finite sum.*

Proof. Take affine open $U = \text{Spec} A$ with $f \in A$. Let $Z = X \setminus U$. Then by our above observation Z only contains finitely many prime divisors. Now for any prime divisor D intersects U , U contains its generic point. So $D \cap U$ is a prime divisor of the open subscheme U . Now $f \in A$ implies f is in the local rings of U , and so $v_D(f) \geq 0$ for those D intersects U , and hence $v_D(f) < 0$ for only finitely many D . Replace f with $1/f$, we see that for all but finitely many D , $v_D(f) = 0$. $\square$

Divisors of the form $\text{div}(f)$ for some $f \in K(X)^\times$ are called **principal divisors**. Two divisors D, D' are **linearly equivalent** if $D - D'$ is principal, denoted by $D \sim D'$. The **class group** $Cl(X) = \text{Div}(X) / \text{Prin}(X)$.

Theorem 5.1. *Let A be a Noetherian integral domain. Then A is a UFD if and only if $X = \text{Spec} A$ is normal, that is A is integrally closed, and $Cl(X) = 0$.*

The proof is actually mildly fun and kind of a revision of commutative algebra. A classic criterion for UFD is that its height-1 primes are principal for noetherian integral domain. And for integrally closed Noetherian domain, the intersection of height-1 local rings is A . This agrees with the result in algebraic number theory.

Theorem 5.2. *Let U open and $Z = X \setminus U$. Then*

1. *There exists a surjective morphism $Cl(X) \rightarrow Cl(U)$, by $[D] \mapsto [D \cap U]$.*
2. *If Z is of codimension at least 2, then this is an isomorphism*
3. *If Z is irreducible and of codimension 1, i.e. a prime divisor, then we have an exact sequence*

$$\mathbb{Z} \xrightarrow{1 \mapsto [Z]} Cl(X) \rightarrow Cl(U) \rightarrow 0$$

The morphism is well-defined because of the observation above and $K(X) = K(U)$. It is surjective since $\overline{D}$ for prime divisor $D \subset U$ is a prime divisor in X and $\overline{D} \cap U = D$. If Z is of codimension at least 2, then it contains no prime divisors, and so no prime divisors are mapped to 0. If Z is a prime divisor, we see that it is the only prime divisor mapped to 0, and so the kernel in $Cl(X)$ is generated by $[Z]$, and the exactness follows. In general, for Z a closed set, we have a s.e.s $\bigoplus_{D \subset Z \text{ prime divisor}} \mathbb{Z}[D] \rightarrow Cl(X) \rightarrow Cl(U) \rightarrow 0$

Example 14. We compute the class group of $= \text{Spec}(k[x, y, z]/(xy - z^3))$. To apply theorem 5.2, we would like to take a prime divisor. We choose $Z = V(x, z) = V(x)$ since $z = 0$ when $x = 0$. Then $U = X \setminus V(x) = D(x) \cong \text{Spec}(k[x, y, z]_x/(xy - z^3))$. Note that $y = x^{-1}z^3$ since x is invertible now. So we can eliminate y and get $U \cong \text{Spec}(k[x, 1/x, z])$ where $k[x, 1/x, z]$ is a noetherian UFD. So $Cl(U) = 0$ and we get an exact sequence $\mathbb{Z} \xrightarrow{1 \mapsto [Z]} Cl(X) \rightarrow 0$. The generic point of Z is $m = (x, z)$. Then $\mathcal{O}_{X, m} = k[x, y, z]_{(x, z)}/(xy - z^3)$. Now $y \notin (x, z)$ is invertible. So similarly we can cancel x and get $\mathcal{O}_{X, m} \cong k[y, z]_{(z)}$. So the maximal ideal is generated by z and so z is a uniformizer. Also y is invertible so $v_L(y) = 0$. Then $v_L(x) = v_L(y^{-1}z^3) = 3$. So $\text{div}(x) = 3Z$ and hence $3[Z] = 0$ in $Cl(X)$.

To show $Cl(X) \cong \mathbb{Z}/3\mathbb{Z}$, it remains to show that $m = (x, z)$ is not principal. Now we look at $(x, y, z) = m'$. Note that m'/m'^2 is a k -vector space with basis $\overline{x}, \overline{y}, \overline{z}$, and the image of m in m'/m'^2 contains $\overline{x}, \overline{z}$ has hence can't be principal.

Example 15. Let A be a Noetherian integral domain, $x \in A$ an element such that the ideal (x) is prime. We want to show that if A_x is UFD, then so is A . By theorem 5.1, this is equivalent to show that A is integrally closed and has $Cl(\text{Spec} A) = 0$. Theorem gives us an exact sequence $\mathbb{Z} \xrightarrow{1 \mapsto D} Cl(\text{Spec}(A)) \rightarrow Cl(D(x) \cong \text{Spec}(A_x)) = 0 \rightarrow 0$ by assumption where $D = V(x)$. Also the generic point of D is (x) which is principal, so the first arrow is in fact a zero map. So $Cl(\text{Spec}(A)) = 0$.

Now given $u \in \text{Frac}(A)$ integral over A , then it is integral over A_x and hence $u \in A_x$. Suppose $u = a/x^n$ with $n \geq 0$ minimal. If $n \geq 1$, let $f(t) = t^m + \sum a_i t^i$ be the integral polynomial such that $f(u) = 0$. Then $u^m = a^m/x^{mn} = S/x^{(m-1)n}$ for some $S \in A$. Then $a^m = Sx \in (x) \implies a \in (x)$. So $a = a'x$ and hence $u = a'/x^{n-1}$, contradicts the minimality of n . So $n = 0$ and $u = a \in A$, and A is integrally closed.

Example 16. We compute $Cl(X = \mathbb{P}^1 \times \mathbb{P}^1)$. Note that $\mathbb{P}^1 \times \mathbb{P}^1 - \mathbb{A}^1 \times \mathbb{A}^1 = \{\infty\} \times \mathbb{P}^1 \cup \mathbb{P}^1 \times \{\infty\}$, where $\{\infty\}$ is the unique point in $\mathbb{P}^1 \setminus \mathbb{A}^1$. Since $k[x, y]$ is a noetherian UFD, $Cl(\mathbb{A}^1 \times \mathbb{A}^1) = Cl(\mathbb{A}^2) = 0$. Then the s.e.s gives a surjection $\mathbb{Z}^2 \rightarrow Cl(\mathbb{P}^1 \times \mathbb{P}^1)$ given by $(1, 0) \mapsto [\ell] = [\{\infty\} \times \mathbb{P}^1]$ and $(0, 1) \mapsto$

$[\ell'] = [\mathbb{P}^1 \times \{\infty\}]$. Now given $f \in K(X) = K(U = \mathbb{A}^1 \times \mathbb{A}^1) = k(x, y)$ with $\text{div}(f) = a[\ell] + b[\ell']$, $\text{div}(f)|_U = 0 \implies f \in \mathcal{O}_U^* \implies f \in k$. Then $\text{div}(f) = 0$ and so $a, b = 0$. So there is no relation between $[\ell]$ and $[\ell']$ in $Cl(X)$ and hence $Cl(X) \cong \mathbb{Z}^2$

Example 17. We compute $Cl(X)$ with $X = k[x, y, z, w]/(xy - zw)$. Pick $V_1 = V(x, z)$ and $V_2 = V(x, w)$. Then $U = X \setminus V_1 \cup V_2 = D(x) \cong \text{Spec}(k[x, y, z, w]_x/(xy - zw)) \cong \text{Spec}(k[x, 1/x, z, w])$. And so $Cl(U) = 0$. So we have a surjective map $\phi : \mathbb{Z}^2 \rightarrow Cl(X)$ given by $(1, 0) \mapsto [V_1]$ and $(0, 1) \mapsto [V_2]$. Now given $f \in K(X)$, suppose $\text{div}(f) = a[V_1] + b[V_2]$. Then $\text{div}(f)|_U = 0$ implies that f a regular invertible function on $U \cong \text{Spec}(k[x, 1/x, z, w])$ and so f is an invertible element in $k[x, 1/x, z, w]$ and so $f = cx^n$ for some $c \in k$ and $n \in \mathbb{Z}$. Now we compute $\text{div}(x)$. Note that $k[x, y, z, w]_{(x, z)}/(xy - zw) = k[x, y, w]_{(x)}$. So x is a uniformizer at $\mathcal{O}_{X, (x, z)}$. Similarly, x is also a uniformizer for the local ring at the generic point of V_2 . So $\text{div}(x) = [V_1] + [V_2]$, and so $\ker \phi = \mathbb{Z}(1, 1)$ and $Cl(X) \cong \mathbb{Z}$

5.2 Cartier Divisors

For $\mathcal{K}_X$ the sheaf of rational functions on X , a **Cartier divisor** is a global section of $\mathcal{K}_X^*/\mathcal{O}_X^*$.

This definition is more than confusing in the first place. First we need to understand $\mathcal{K}_X$. Morally, it is the “largest possible fraction” $\mathcal{O}_X$ can get. If X is integral, this is nothing other than the constant sheaf of $K(X)$.

Now by the definition of quotient sheaf, a Cartier divisor can also be given by the data $\{(U_i, f_i)\}$ where U_i forms a covering of X and f_i is a rational function on U_i , such that $f_i/f_j \in \mathcal{O}_X^*(U_i \cap U_j)$, i.e. f_i/f_j is a unit on the intersection.

Now we resume to assumption in section 5.1. Then given a Cartier divisor $\mathcal{D}$ above, $f_i \in K(X)^*$, and we define a Weil divisor $\sum n_D D$ with $n_D = v_D(f_i)$ where D intersects U_i . This is Well defined because on the intersection f_i/f_j is a unit, so $v_D(f_i) = v_D(f_j)$ for prime divisor D in the intersection.

Note that by our definition, $\sum n_D D|_{U_i} = \text{div}(f_i)|_{U_i}$. This is why we can think of Cartier divisors as **locally principal** divisors.

Proposition 5.2. *If the local rings of X are UFD, the Weil divisors are Cartier, and hence $CaCl(X) = Cl(X)$*

This condition is called **locally factorial**. Smoothness implies locally factorial, because regular local rings are UFD.

Given a Cartier divisor D with data $\{(U_i, f_i)\}$, we can define an associated line bundle $\mathcal{O}_X(D)$ by letting $\mathcal{O}_X(D)(U_i)$ be the $\mathcal{O}_X$ -submodule generated by f_i^{-1} . This is well-defined because $\mathcal{O}_{U_i \cap U_j} f_i^{-1} = \mathcal{O}_{U_i \cap U_j} f_j^{-1}$. An equivalent definition is to let $\mathcal{O}_X(D)(U_i)$ consist of those sections f in $K(X)^*$ such that $(\text{div}(f_i) + D)|_{U_i}$ effective.

Conversely, given a line bundle, we can recover a Cartier divisor if X is integral. Also it is not hard to see that D is principal if and only if $\mathcal{O}_X(D) \cong \mathcal{O}_X$. Hence $CaCl(X) \cong Pic(X)$ for an integral scheme X .

6 Sheaf Cohomology

Recall that the functor $\Gamma(X, -)$ is left exact, i.e. given a short exact sequence of sheaves $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$, we have an exact sequence $0 \rightarrow \Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'')$.

Just as every (co)homology theory, sheaf cohomology is to evaluate the failure of exactness.

Recall the definition of injective modules. Analogously, we say a sheaf $\mathcal{I}$ is **injective** if for a diagram of sheaves

$$\begin{array}{ccc} & \mathcal{I} & \\ \uparrow & \nwarrow & \\ A & \longrightarrow & B \end{array}$$

there exists a morphism $\mathcal{B} \rightarrow \mathcal{I}$ to make this commute. We can memorize this as any morphism $\mathcal{A} \rightarrow \mathcal{I}$ can extend to a morphism $\mathcal{B} \rightarrow \mathcal{I}$.

Recall that we say a category **has enough injectives** if any objects can be embedded into an injective object.

Proposition 6.1. *The category of sheaves of abelian groups has enough injectives.*

The proof is similar to that for the category of abelian groups and modules, and we will not be bothered by this.

Therefore, any sheaf of abelian groups admits an **injective resolution**, i.e. an exact sequence $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}_0 \rightarrow \dots$ where $\mathcal{I}_i$ are injective sheaves. Then idea is to embed the cokernel at each step into a new injective sheaf. The injective resolution is denoted by $\mathcal{F} \rightarrow \mathcal{I}_\bullet$.

Apply the functor $\Gamma(X, -)$ to $\mathcal{I}_\bullet$, we get a chain complex $0 \rightarrow \Gamma(X, \mathcal{I}_0) \rightarrow \Gamma(X, \mathcal{I}_1) \rightarrow \dots$. The **i-th sheaf cohomology group** of $\mathcal{F}$ is

$$H^i(X, \mathcal{F}) = \ker(\Gamma(X, \mathcal{I}_i) \rightarrow \Gamma(X, \mathcal{I}_{i+1})) / \text{Im}(\Gamma(X, \mathcal{I}_{i-1}) \rightarrow \Gamma(X, \mathcal{I}_i))$$

Just as any left-exact functor, we have the following results for sheaf cohomology

Theorem 6.1. 1. $H^0(X, \mathcal{F}) = \Gamma(X, \mathcal{F})$

2. $H^i(X, \mathcal{F})$ is independent of the choice of injective resolution $\mathcal{I}_\bullet$ of $\mathcal{F}$

3. Any short exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ induces a long exact sequence of their cohomology groups

Theorem 6.2. (Grothendieck) Let X be a Noetherian topological space of dimension n and $\mathcal{F}$ a sheaf of abelian groups on X . Then $H^i(X, \mathcal{F}) = 0$ for $i > n$.

We say a sheaf is **acyclic** if $H^i(X, \mathcal{F}) = 0$ for $i > 1$. Then injective sheaves are acyclic, by considering its trivial resolution. We can compute sheaf cohomology via acyclic resolution:

Proposition 6.2. For $\mathcal{F}$ a sheaf on X with a resolution by acyclic sheaves: $0 \rightarrow \mathcal{F} \rightarrow \mathcal{F}_0 \xrightarrow{d^0} \dots$. Then $H^i(X, \mathcal{F}) = \ker(d^i : \Gamma(X, \mathcal{F}_i) \rightarrow \Gamma(X, \mathcal{F}_{i+1})) / \text{Im}(d^{i-1} : \Gamma(X, \mathcal{F}_{i-1}) \rightarrow \Gamma(X, \mathcal{F}_i))$

We can prove this by splitting the resolution into short exact sequences for each $\mathcal{F}_i$ and then take the long exact sequence of sheaf cohomology. After recollecting the data we see this follows.

A sheaf is called **flasque** if the restriction maps are surjective. An important fact of flasque sheaves is the following

Proposition 6.3. Given a short exact sequence of sheaves $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$, if $\mathcal{F}'$ is flasque, then we have an exact sequence $0 \rightarrow \Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'') \rightarrow 0$

The proof is by gluing sections, with the idea similar to a lemma of showing quasi-coherent sheaves are locally affine.

Corollary 6.1. *Flasque sheaves are acyclic*

See [Har77, Chapter III proposition 2.5]. This implies that we can use flasque resolutions to compute cohomology.

Example 18. The sheaf of rational functions $\mathcal{K}_X$ is flasque if X is an integral scheme. In this case $\mathcal{K}_X$ is just the constant sheaf $\overline{K(X)}$, and it is clear that $\mathcal{K}_X$ is flasque.

Why is it important? We have a natural short exact sequence $0 \rightarrow \mathcal{O}_X^* \rightarrow \mathcal{K}_X^* \rightarrow \mathcal{K}_X^*/\mathcal{O}_X^* \rightarrow 0$, which induces a long exact sequence. We have an exact sequence

$$K(X)^* \rightarrow \Gamma(X, \mathcal{K}_X^*/\mathcal{O}_X^*) \rightarrow H^1(X, \mathcal{O}_X^*) \rightarrow 0$$

$\Gamma(X, \mathcal{K}_X^*/\mathcal{O}_X^*)$ is exactly the Cartier divisors on X , while the image of $\Gamma(X, \mathcal{K}_X^*)$ in $\Gamma(X, \mathcal{K}_X^*/\mathcal{O}_X^*)$ is exactly the principal divisors. So we get $\text{CaCl}(X) \cong H^1(X, \mathcal{O}_X^*)$

6.1 Čech Cohomology

In practice it is almost impossible to compute cohomology through injective resolution. Instead, we have the **Čech cohomology**, which is analogous to singular homology in algebraic topology.

Let $\mathcal{F}$ be a sheaf on X and $\mathcal{U} = \{U_i\}_{i \in I}$ be an open covering with I an well-ordered set. Write $U_{i_0 \dots i_k}$ for $U_{i_0} \cap \dots \cap U_{i_k}$

The group of **Čech n-cochains** is $C^n(\mathcal{U}, \mathcal{F}) = \prod_{i_0 < \dots < i_n} \mathcal{F}(U_{i_0 \dots i_n})$

We can define a coboundary map $d = d^n : C^n(\mathcal{U}, \mathcal{F}) \rightarrow C^{n+1}(\mathcal{U}, \mathcal{F})$. For $\alpha \in C^n(\mathcal{U}, \mathcal{F})$, we define

$$(d\alpha)_{i_0 \dots i_{n+1}} = \sum_{k=0}^{n+1} (-1)^k \alpha_{i_0 \dots \widehat{i_k} \dots i_{n+1}}|_{U_{i_0 \dots i_{n+1}}}$$

which is analogous to the boundary map of the singular complex. Therefore, one easily tell see that $d \circ d = 0$. So we get a cochain complex, known as **Čech complex**. The **i-th Čech cohomology** of the pair $(\mathcal{U}, \mathcal{F})$ is $\ker d_i / \text{imd}_{i-1}$

To compute Čech cohomology, we need to find a nice and natural open covering $\mathcal{U}$.

Example 19. Take $X = S^1$, $U = S^1 - \{1\}$, $V = S^1 - \{-1\}$ and $\mathcal{F} = \underline{\mathbb{Z}}$ the consteant sheaf. Then $C^0(\mathcal{U}, \mathcal{F}) = \mathbb{Z}^2$, $C^1(\mathcal{U}, \mathcal{F}) = \mathbb{Z}^2$, and d_0 takes (a, b) to $(b-a, b-a)$. We see that $\check{H}^0(\mathcal{U}, \mathcal{F}) = \check{H}^1(\mathcal{C}, \mathcal{F}) = \mathbb{Z}$.

Proposition 6.4. *Let X be a Noetherian scheme with an open affine cover $\mathcal{U} = \{U_i\}$ with the property that $U_{i_0 \dots i_n}$'s are all affine (e.g. when X is separated). Then if $\mathcal{F}$ is a quasi-coherent sheaf of $\mathcal{O}_X$ -module, then $\check{H}^i(\mathcal{U}, \mathcal{F}) \cong H^i(X, \mathcal{F})$*

So when the covering and its intersections are affine, we can use the Čech cohomology to compute sheaf cohomology. In particular, $H^i(\mathcal{U}, \mathcal{F}) = 0$ for $i > 0$ if X is the spectrum of a noetherian ring, since it is an affine cover of itself

Example 20. Recall the **punctured plane** $X = \mathbb{A}_k^2 \setminus (0, 0)$, an open subscheme of $\mathbb{A}_k^2$. If it is affine, then Hartshorne III 3.5 tells us that since it is affine over k , its i-th sheaf cohomology vanishes for $i > 0$ and any quasi-coherent sheaf $\mathcal{F}$. On the other hand, it admits a covering by $D(x)$ and $D(y)$. Then $\mathcal{O}_X(D(x)) = k[x, 1/x, y]$ and so on. Hence $H^1(X, \mathcal{O}_X) = \check{H}^1(X, \mathcal{O}_X) = k[x, 1/x, y, 1/y]/(k[x, 1/x, y] + k[x, y, 1/y])$ by proposition 6.3. We see the k -basis are $1/x^a y^b$ with $a, b > 0$. In particular, this group is of infinite dimension over k , which is a contradiction.

For projective space, their affine charts form a natural candidate for the covering to choose.

We consider $\mathcal{O}_{\mathbb{P}^1}(-2)$. For the affine cover $D_+(x_0), D_+(x_1)$ of $\mathbb{P}_k^1$, $\mathcal{O}_{\mathbb{P}^1}(-2)(D_+(x_0)) = S(-2)_{(x_0)}$ which is generated by $1/x_0^2, x_1/x_0^3, \dots$. Similarly for $\mathcal{O}_{\mathbb{P}^1}(-2)(D_+(x_1))$, while $\mathcal{O}_{\mathbb{P}^1}(-2)(D_+(x_0) \cap D_+(x_1)) = \mathcal{O}_{\mathbb{P}^1}(-2)(D_+(x_0 x_1))$ is generated by $1/(x_0 x_1)$ and the generators of $\mathcal{O}_{\mathbb{P}^1}(-2)(D_+(x_0))$ and $\mathcal{O}_{\mathbb{P}^1}(-2)(D_+(x_1))$. Hence $H^1(\mathbb{P}_k^1, \mathcal{O}_{\mathbb{P}^1}(-2)) = k$ by proposition 6.4. More generally, we have the following theorem.

Theorem 6.3. *Let $S = k[x_0, \dots, x_r]$, then*

1. $S \cong \bigoplus_{n \geq 0} H^0(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(n))$
2. $H^i(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(n)) = 0$ for $0 < i < r$ and any n
3. $H^r(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(-r-1)) = k$
4. *there is a perfect pairing $H^0(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(n)) \times H^r(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(-n-r-1)) \rightarrow H^r(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(-r-1)) = k$*

(1) is true because $\mathcal{O}_{\mathbb{P}^r}(n)$ is globally generated by homogeneous degree- n polynomials. (2) is in fact the most difficult one, by fiddling with a short exact sequence. (3) is a generalization of the argument for $\mathcal{O}_{\mathbb{P}^1}(-2)$. (4) follows from (3).

Remark 6.1. (4) is the first form of Serre duality that we saw.

6.2 Exercise

Proposition 6.5. *Let X be a subscheme of $\mathbb{P}_k^2$ defined by a single homogeneous polynomial $f(x_0, x_1, x_2) = 0$. Assume that $(1, 0, 0) \notin X$. Then show X can be covered by the two affine open subsets $U = X \cap D_+(x_1)$, $V = X \cap D_+(x_2)$. Compute the Čech cohomology.*

7 Differentials and Riemann-Roch

The motivation for Kahler differentials, again, comes from commutative algebra. Let B be an A -algebra. For a B -module M , an **A-derivation** $d : B \rightarrow M$ is an A -linear homomorphism satisfying the Leibniz rule. The **module of relative differentials** $\Omega_{B/A}$ is a B -module equipped with an A -linear derivation $d : B \rightarrow \Omega_{B/A}$ satisfying the following universal property: for any B -module M and any A -linear derivation $d' : B \rightarrow M$, d' factors uniquely through d via a B -homomorphism $\Omega_{B/A} \rightarrow M$.

There are two constructions for $\Omega_{B/A}$. The formal one is to consider the B -module generated by symbols db for $b \in B$, modulo the relations from derivations. The concrete one is from the diagonal map $B \otimes_A B \rightarrow B$, $b_1 \otimes b_2 \mapsto b_1 b_2$. Let I be the kernel of the diagonal map. Then one can show that I/I^2 equipped with map $d(b) = b \otimes 1 - 1 \otimes b$ (note that I/I^2 is a $(B \otimes B/I) \cong B$ -module) has the universal property.

Now consider a separated morphism $f : X \rightarrow Y$. Let $\mathcal{I}$ be the defining ideal of the closed immersion $\Delta : X \rightarrow X \times_Y X$. We define the **sheaf of Kahler differentials**

$$\Omega_{X/Y} = \Delta^*(\mathcal{I}/\mathcal{I}^2)$$

Remark 7.1. If f is not separated, Δ is still a **general immersion**, i.e. factors through some open subscheme U with $X \rightarrow U$ closed immersion. Then Δ gives rise to a sheaf of ideals on an open subscheme containing the image of Δ , and we may then proceed as in the separated case.

A good reference for Kahler differentials and unramified morphism is [BLR90, Chapter II]

7.1 Riemann-Roch

Suppose X is a smooth variety over k . Then $\Omega_{X/k} = \Omega_X$ is locally free of order $\dim X$. Then we define the **canonical sheaf** on X $\omega_X = \wedge^{\dim X} \Omega_X$ which is a line bundle. We call its corresponding Cartier divisor class K_X the **canonical divisor** of X .

The canonical sheaf is important in the study of varieties. Now we see its presence in Riemann-Roch and Serre-duality.

Theorem 7.1. (*Serre duality*) Let X be a smooth projective variety over k of dimension n . For any locally free sheaf $\mathcal{F}$ on X of finite order, there is a natural isomorphism

$$H^i(X, \mathcal{F}^\vee \otimes \omega_X) \xrightarrow{\sim} H^{n-i}(X, \mathcal{F})^\vee$$

In general, for a projective scheme X over k and coherent sheaf $\mathcal{F}$ on X , $H^i(X, \mathcal{F})$ is of finite dimension over k . We define the **Euler characteristic** of $\mathcal{F}$ to be

$$\chi(\mathcal{F}) = \sum_{i=0}^{\dim X} (-1)^i \dim_k(H^i(X, \mathcal{F}))$$

which is additive on exact sequences.

Now let X a smooth projective curve over an algebraically closed k . The closed points on X are exactly the prime divisors. So for $D \in \text{Div} X$, $D = \sum a_i P_i$. Let $\deg(D) = \sum a_i$. We will use without proof that $\chi(\mathcal{O}_X(D)) = \chi(\mathcal{O}_X) + \deg(D)$. Let $g = \dim(H^1(X, \mathcal{O}_X))$ be the **genus** of X .

Theorem 7.2. (*Riemann-Roch*) For $D \in \text{Div} X$,

$$\dim(H^0(X, \mathcal{O}_X(D))) - \dim(H^0(X, \omega_X \otimes \mathcal{O}_X(-D))) = \deg(D) + 1 - g$$

Proof. Note that $\chi(\mathcal{O}_X(D)) = \dim(H^0(X, \mathcal{O}_X(D))) - \dim(H^1(X, \mathcal{O}_X(D))) = \dim(H^0(X, \mathcal{O}_X(D))) - \dim(H^0(X, \omega_X \otimes \mathcal{O}_X(-D)))$ by Serre-duality. But $\chi(\mathcal{O}_X(D)) = \chi(\mathcal{O}_X) + \deg(D) = 1 - g + \deg(D)$. Recall that the global sections of a projective variety over algebraically closed k is just the constant. $\square$

Remark 7.2. Notice that Serre-duality gives $\chi(\omega_X) = -\chi(\mathcal{O}_X) = g - 1$. Replacing D with $\mathcal{K}_X$ in Riemann-Roch, we get $\deg(\mathcal{K}_X) = 2g - 2$.

Remark 7.3. If $\deg D < 0$, then $H^0(X, \mathcal{O}_X(D)) = 0$, since $\deg(\text{div}(f) + D) = \deg D < 0$ for any $f \in K(X)^*$. If $\deg D > 2g - 2$, then $\deg(\mathcal{K}_X - D) < 0$ from remark 7.2 and so $H^0(X, \omega_X \otimes \mathcal{O}_X(-D)) = 0$. Then Riemann-Roch gives $\dim(H^0(X, \mathcal{O}_X(D))) = \deg D + 1 - g$

Using Riemann-Roch, we can show that D is very ample if $\deg D > 2g$ and D is basepoint-free if $\deg D > 2g - 1$.

References

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